



Emergence, Onset Timing, and Predictors of Probable Insomnia in Previously Low-Symptom Pregnant Individuals Across Pregnancy and Postpartum



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BACKGROUND

Perinatal insomnia is associated with adverse mental health and birth outcomes. However, little is known about who is most at risk.

We examined the **emergence** and **course of probable insomnia** from late pregnancy to one year postpartum among those with low symptoms (< 8 Insomnia Severity Index [ISI]) in late pregnancy (26-32 weeks).

SAMPLE & METHOD

Participants: 321 pregnant women (Age: 33.6 ± 3.8 years) with low insomnia symptoms at late pregnancy.

Outcome: Probable insomnia (categorised as $ISI \geq 10$) assessed at baseline, 5 weeks later, and 3, 6 and 12 months postpartum.

Predictors:



Maternal Age



Existing Children (Parity)



Education (Above bachelor vs less than bachelor)



Dysfunctional Beliefs About Sleep



Sleep Reactivity (Ford Insomnia Response to Stress Test)

Analysis: Multinomial regressions, using the Never Insomnia group as the reference, examined predictors of trajectory.

TRAJECTORY OF PROBABLE INSOMNIA

Four trajectories were examined (**Figure 1**):

Never Insomnia: never met probable insomnia threshold,

Insomnia Recovery: met threshold in pregnancy and/or early postpartum but recovered by 12-months-postpartum;

Persistent Insomnia: met threshold in pregnancy and/or early postpartum which persisted at 12-months-postpartum;

Late Postpartum-Onset Insomnia: first met threshold at 12-months-postpartum.

MULTINOMIAL REGRESSIONS

Compared to “**Never Insomnia**” (**Figure 2**):

↑ Higher **DYSFUNCTIONAL BELIEFS ABOUT SLEEP** = **Insomnia Recovery**, **Persistent Insomnia** & **Late Postpartum-Onset**.

↑ Higher **SLEEP REACTIVITY** = **Persistent Insomnia** & **Insomnia Recovery**.

Having existing children (**PARITY**) = **Persistent Insomnia**. **AGE** and **EDUCATION** were not associated with any trajectory.

CONCLUSION

Among women with low insomnia symptoms in late pregnancy, distinct insomnia trajectories emerge.

Dysfunctional sleep beliefs and sleep reactivity may be key risk factors for insomnia development.

Having existing children may increase vulnerability to persistent symptoms.

Targeting maladaptive sleep beliefs, and identifying individuals with heightened sleep reactivity may offer opportunities for early prevention of perinatal insomnia.

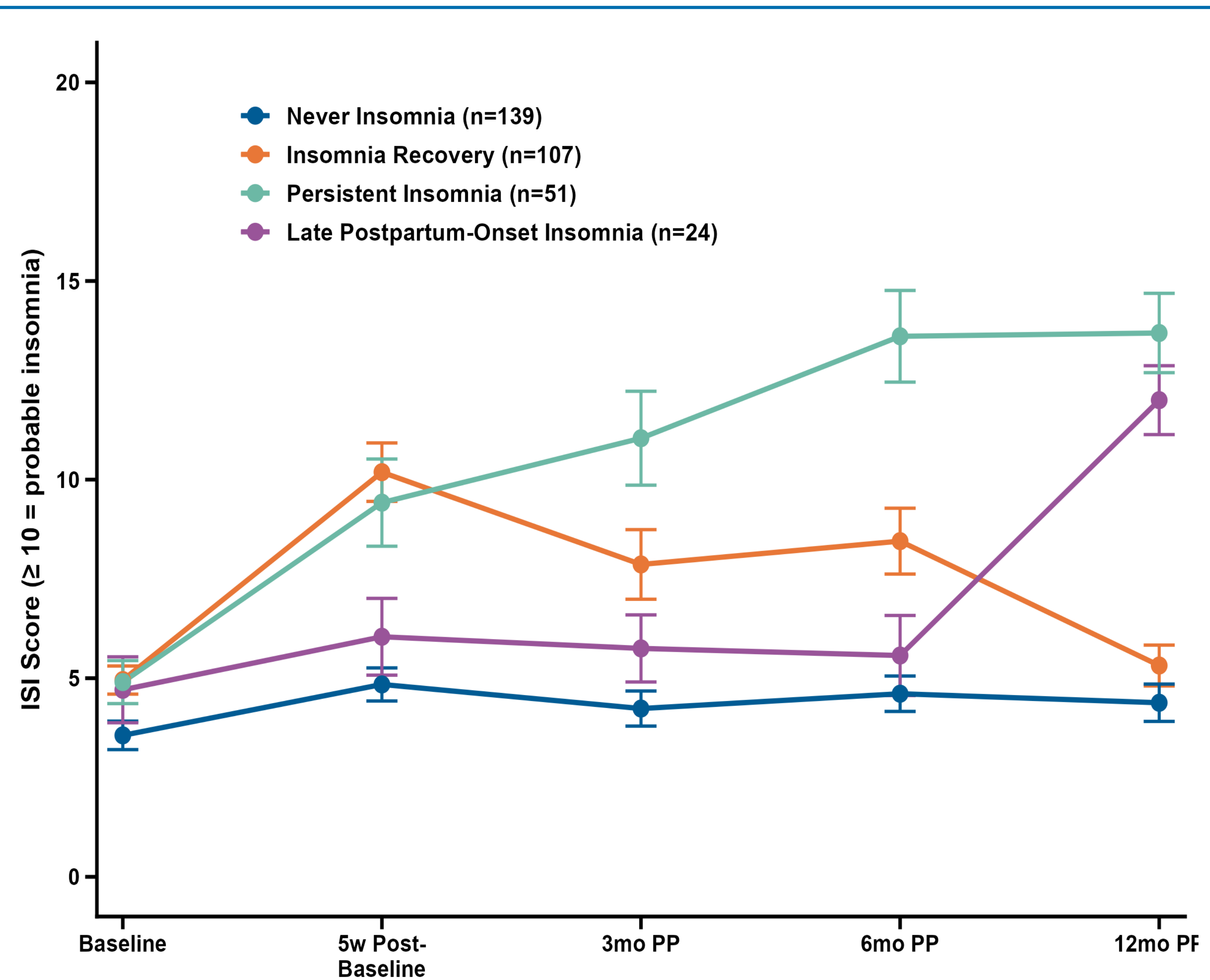
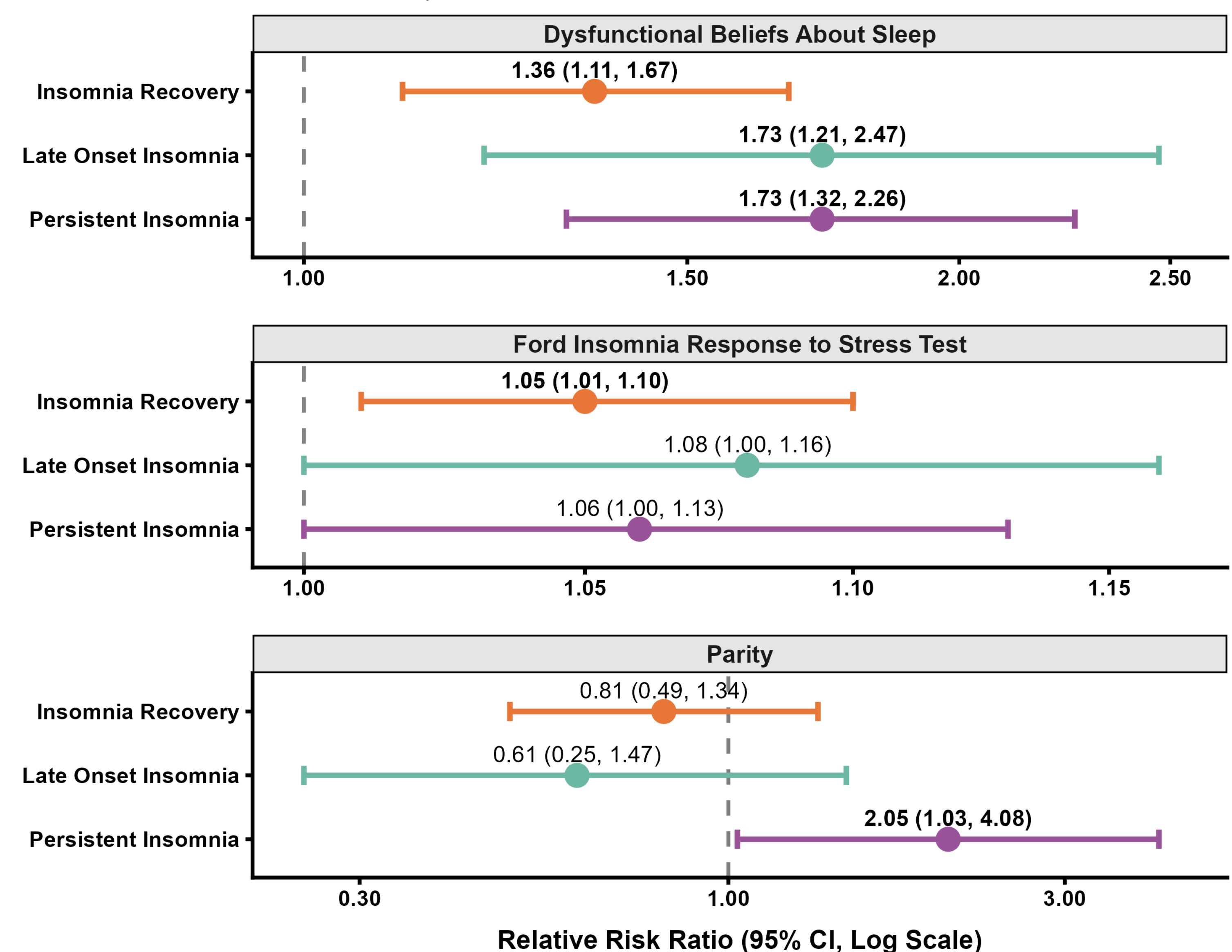


Figure 1. Trajectory of insomnia symptoms (ISI) from late pregnancy (26-32 weeks) to 12 months postpartum, colour coded by identified trajectory of probable insomnia.

Relative Risk Ratios of Probable Insomnia

Reference Group: Never Insomnia



Note: Bold numbers indicate statistically significant estimates.

Figure 2. Forest plot of the Relative Risk Ratio & 95% Confidence Intervals for Multinomial Regressions of predictors of insomnia trajectory relative to “Never Insomnia”